

NOTES FOR ARBEITSGEMEINSCHAFT: NEW HOMOLOGICAL METHODS IN TORIC GEOMETRY

MICHAEL K. BROWN, DAVID EISENBUD, AND DANIEL ERMAN

1. GOOD MULTIGRADINGS

We let $S = k[x_1, \dots, x_n]$ be a polynomial ring and we will use $\text{Cl}(S)$ to denote an arbitrary finitely generated abelian group. We will consider S as graded with respect to $\text{Cl}(S)$ via a map $\mathbb{Z}^n \xrightarrow{\deg} \text{Cl}(S)$ where $\varepsilon_1, \dots, \varepsilon_n$ are a fixed basis of $\mathbb{Z}^n$ and where we set $\deg(x_i) := \deg(\varepsilon_i) \in \text{Cl}(S)$. To avoid trivialities, we will always assume that the map $\deg$ is surjective.

Our goal in this section is to articulate conditions under which the multigraded ring S corresponds to the Cox ring of some projective toric variety.

We will need some notation. We write $M := \ker(\deg)$ and we consider the induced short exact sequence:

$$0 \rightarrow M \xrightarrow{\alpha} \mathbb{Z}^n \xrightarrow{\deg} \text{Cl}(S) \rightarrow 0.$$

We will define $\text{Eff}(S) := \{d \in \text{Cl}(S) \mid S_d \neq 0\}$. We refer to this as the **effective semigroup** as it forms a subsemigroup of $\text{Cl}(S)$. We can consider the corresponding cone $\text{Eff}(S)_{\mathbb{Q}} \subseteq \text{Cl}(S) \otimes_{\mathbb{Z}} \mathbb{Q}$ which is the positive rational cone spanned by the image of $\text{Eff}(S)$ under the natural map to $\text{Cl}(S) \otimes_{\mathbb{Z}} \mathbb{Q}$.

Definition 1.1. We say that S is a **positively graded ring** if there exists a group homomorphism $\varphi: \text{Cl}(S) \rightarrow \mathbb{Z}$ such that $\varphi \circ \deg(\varepsilon_i) > 0$ for all i .

Lemma 1.2. *The following are equivalent:*

- (1) S is positively graded.
- (2) $S_0 = k$.
- (3) $\text{Eff}(S)_{\mathbb{Q}}$ is a pointed cone.

Proof. (1) implies (2) is easy.

For (2) implies (3): If $\text{Eff}(S)_{\mathbb{Q}}$ is not a pointed cone then there exist $d \in \text{Cl}(S)$ where d and $-d$ both lie in $\text{Eff}(S)$ and then $S_0 \neq k$. (Details to add, especially in torsion case.)

For (3) implies (1): Define the positive grading φ by choosing a vector space $\text{Cl}(S) \otimes \mathbb{Q} \rightarrow \mathbb{Q}$ that sends $\text{Eff}(S)_{\mathbb{Q}}$ into $\mathbb{Q}_{>0}$ and then clearing denominators as needed $\square$

Proposition 1.3. *The following are equivalent:*

- (1) *There is a polynomial ring S , graded via $\deg: \mathbb{Z}^n \rightarrow \text{Cl}(S)$, that satisfies the following conditions:*
 - (a) $\deg$ is surjective;
 - (b) S is positively graded;
 - (c) Each row of α is the primitive point on a distinct ray.
- (2) *There is a simplicial, projective toric variety X with $\text{Cl}(X) = \text{Cl}(S)$ and where the Cox ring of X equals S .*

Proof. (2) implies (1) is easy. Given such an X we start choose a basis for the lattice N and take the corresponding basis for the dual lattice M . Since the Cox ring of X is S , we know that X involves n rays, and so we have the matrix of rays $N \leftarrow \mathbb{Z}^n$. Since the rays are distinct (and nonzero) the columns of this matrix all nonzero and are not scalar multiples of one another. Dualizing this

then yields a map $\alpha: M \rightarrow \mathbb{Z}^n$ satisfying condition (iii). We set $\text{Cl}(S) := \text{Cl}(X)$ and the induced grading satisfies (i) and (ii) by standard facts.

(1) implies (2) is more subtle. The $\text{Cl}(S)$ -grading on S induces a GIT problem and hence a secondary fan. If X_Σ is the toric variety associated to some of the secondary fan, Proposition 14.4.1 (a basic fact on Gale duality) implies that the support of the fan Σ is the cone spanned by the rays in $N_\mathbb{Q}$. Lemma 14.3.2 then implies that if $\text{Eff}(S)$ is pointed (C_β in the CLS notation), then the cone spanned by the rays (C_ν in CLS notation) is all of $N_\mathbb{Q}$. In particular, any X_Σ from the secondary fan will be proper.

Proposition 14.4.1 also implies that each X_Σ is semiprojective and since it is proper by the previous paragraph, it is also projective.

We next utilize Proposition 15.1.6 of CLS. This says that we have a chamber of the secondary fan where the corresponding variety X_Σ requires n rays if and only if the rows of α are nonzero and generate distinct rays, which is implied by condition (c). In this case, $\alpha^*: \mathbb{Z}^n \rightarrow N$ will determine the rays of X_Σ . The stronger condition, that the rows of α each represent a primitive point on a distinct ray, then implies that the ray matrix for X_Σ will be precisely α^* . Since the Class group of X is computed in terms of the ray matrix, we have that the class group of X will equal $\text{Cl}(S)$, and the map $\mathbb{Z}^n \rightarrow \text{Cl}(X)$ will agree with the map $\deg$. Thus the Cox ring of X_Σ will be a polynomial ring on n variables and the grading will be as desired. $\square$

2. EQUIVALENT CONDITIONS FOR BONDAL-THOMSEN

The Bondal-Thomsen collection is a finite set of line bundles that can be defined in several different ways. We will take the following definition, originally due to Bondal, as our definition. Consider the short exact sequence

$$0 \rightarrow M \xrightarrow{\alpha} \mathbb{Z}^n \xrightarrow{\deg} \text{Cl}(X) \rightarrow 0.$$

We write α_ρ for the projection of α onto the basis vector corresponding to ρ in $\Sigma(1)$. (Explicitly we have $\alpha_\rho(m) = \langle -m, u_\rho \rangle$ in the standard notation of CLS, but I don't think this is actually so relevant.) Identifying $\mathbb{Z}^n$ with torus invariant divisors, we explicitly have $\alpha(m) = \sum_\rho \alpha_\rho(m) D_\rho$ and the class of $\alpha(m)$ is 0 in $\text{Cl}(X)$ for all $m \in M$.

We now extend this to $M \otimes \mathbb{Q}$ as follows. We write $\alpha \otimes \mathbb{Q}$ for the corresponding map of vector spaces $M \otimes \mathbb{Q} \rightarrow \mathbb{Q}^n$. And we will write $[\alpha]$ for the map

$$[\alpha]: M \otimes \mathbb{Q} \rightarrow \mathbb{Z}^n \text{ where } m \mapsto \sum_\rho \lfloor \alpha_\rho(m) \rfloor D_\rho.$$

We note that, while $\alpha(m)$ will always be trivial as an element of $\text{Cl}(X) \otimes \mathbb{Q}$, this is not the case for $[\alpha](m)$.

Definition 2.1. The Bondal-Thomsen collection $\Theta_X \subseteq \text{Cl}(X)$ is the image of $\deg \circ [\alpha]$.

Note that, since the sublattice $M \subseteq M \otimes \mathbb{Q}$ maps to 0 in $\text{Cl}(X)$, it would be equivalent to consider points in the rational torus $M \otimes \mathbb{Q}/M$.

Example 2.2. Let $X = \mathbb{P}^2$. Then $M = \mathbb{Z}^2$ and α is given by the matrix

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ -1 & -1 \end{pmatrix}$$

Since it suffices to consider points in $M \otimes \mathbb{Q}/M$, it suffices to consider $(a, b) \in M \otimes \mathbb{Q}^2$ where $0 \leq a, b < 1$. In this case we have

$$[\alpha](a, b) = \lfloor -a \rfloor D_0 + \lfloor -b \rfloor D_1 + \lfloor a + b \rfloor D_2.$$

Since all the prime torus divisors D_0, D_1, D_2 are equivalent to $\mathcal{O}_{\mathbb{P}^2}(1)$, determining the Bondal-Thomsen collection for $\mathbb{P}^2$ amounts to answering: which integers can arise as a sum

$$\lfloor -a \rfloor + \lfloor -b \rfloor + \lfloor a + b \rfloor \text{ with } (a, b) \in [0, 1]^2.$$

If $(a, b) = (0, 0)$ we get 0. If $0 < a < 1$ and $b = 0$ or vice versa, we get -1 . $0 < a < 1$ and $0 < b < 1$ and $a + b < 1$ then we get -2 , otherwise we get -1 . In particular, we conclude that the Bondal-Thomsen collection is $\mathcal{O}, \mathcal{O}(-1), \mathcal{O}(-2)$ in this case.

Definition 2.3. The Koszul zonotope is the locus

$$\mathcal{Z} := \sum_{\rho} a_{\rho} \deg(x_{\rho}) \subseteq \text{Cl}(X) \otimes \mathbb{Q}^1 \text{ with } a_{\rho} \in (-1, 0].$$

Definition 2.4. The toric Frobenius map of degree ℓ is the morphism $F_{\ell}: X \rightarrow X$ that is induced by the map of Cox rings $S \rightarrow S$ given by sending $x_{\rho} \mapsto x_{\rho}^{\ell}$ for all ρ .

Example 2.5. Continuing the example of $\mathbb{P}^2$, the zonotope $\mathcal{Z}$ is simply the half-open interval $(-3, 0]$. Thus the lattice points correspond to $-2, -1$ and 0 . Moreover, if we consider

Theorem 2.6. *The follows subsets of $\text{Cl}(X)$ are all equal to the Bondal-Thomsen collection.*

- (1) *The set of degrees $d \in \text{Cl}(X)$ as in Definition 2.1.*
- (2) *The set of degrees $d \in \text{Cl}(X)$ such that: in $\text{Cl}(X) \otimes \mathbb{Q}$, d is equivalent to an element $\sum_{\rho} a_{\rho} D_{\rho}$ with $a_{\rho} \in (-1, 0]$.*
- (3) *The set of degrees $d \in \text{Cl}(X)$ such that $\deg_{\mathbb{Q}}(d)$ lies in the zonotope $\mathcal{Z}$.*
- (4) *The set of degrees $d \in \text{Cl}(X)$ such that: there is some $\ell > 0$, where $\mathcal{O}_X(d)$ appears as a summand of $F_{\ell,*} \mathcal{O}_X$.*
- (5) *The set of degrees $d \in \text{Cl}(X)$ such that: for any $\ell \gg 0$, $\mathcal{O}_X(d)$ appears as a summand of $F_{\ell,*} \mathcal{O}_X$.*

Proof. (1) $\Rightarrow$ (2): Let $m \in M \otimes \mathbb{Q}$. Consider the difference

$$\lfloor \alpha \rfloor(m) - \alpha(m) = \sum_{\rho} (\lfloor \alpha_{\rho}(m) \rfloor - \alpha_{\rho}(m)) D_{\rho}.$$

But for any rational number z , the difference between $\lfloor z \rfloor$ and z lies in $(-1, 0]$ and ehnce this divisor has the desired form. Since $\alpha(m)$ is trivial in $\text{Cl}(X) \otimes \mathbb{Q}$ we have the desired result.

(2) $\iff$ (3) is trivial.

(2) $\iff$ (4) $\iff$ (5) appear in [?achinger, Theorem 2], in the case $D = 0$. $\square$

One important fact about the Bondal-Thomsen collection is the following easy consequence of Demazure Vanishing.

Proposition 2.7. *Let $a \in \text{Cl}(X)$ correspond to an ample class and let $d \in \text{Cl}(X)$ corresponds to an element of the Bondal-Thomsen collection Θ . Then*

$$H^i(X, \mathcal{O}_X(a + d)) = 0 \text{ for all } i > 0.$$

3. CHAMBERS AND IRRELEVANT IDEALS

Fix a monomial $\mathbf{x}^{\mathbf{a}} := x_1^{a_1} \cdots x_n^{a_n}$ and a point $d \in \text{Cl}(S) \otimes \mathbb{Q}$. We say that $\mathbf{x}^{\mathbf{a}}$ spans d if the positive, rational cone spanned by $\{\deg_{\mathbb{Q}}(x_i) | a_i \neq 0\}$ contains d , i.e. if there exist $q_i \in \mathbb{Q}_{>0}$ such that $d = \sum_{a_i \neq 0} q_i \deg_{\mathbb{Q}}(x_i)$. This allows us to define a correspondence between convex cones in $\text{Cl}(S) \otimes \mathbb{Q}$ and squarefree monomial ideals in S . We note that, because we insist that the coefficients q_i be strictly positive, this is an open condition on the points d .

¹As a technical note: when $\text{Cl}(X)$ has torsion, we always consider degrees under the natural map $\text{Cl}(X) \rightarrow \text{Cl}(X) \otimes \mathbb{Q}$.

Definition 3.1. For a convex cone $\Gamma \subseteq \text{Cl}(X) \otimes \mathbb{Q}$ we define

$$\text{ideal}(\Gamma) := \langle \mathbf{x}^{\mathbf{a}} \mid \mathbf{x}^{\mathbf{a}} \text{ spans } d, \forall d \in \Gamma \rangle.$$

Conversely, given a squarefree monomial ideal B we define

$$\text{cone}(B) := \{d \mid \mathbf{x}^{\mathbf{a}} \text{ spans } d, \forall \mathbf{x}^{\mathbf{a}} \in B\}.$$

Observe that $\text{ideal}(\Gamma)$ is a squarefree monomial ideal in S , and $\text{cone}(B)$ is an open convex cone in $\text{Cl}(S) \otimes \mathbb{Q}$. Moreover, it is elementary to check that

$$\text{ideal}(\text{cone}(\text{ideal}(\Gamma))) = \text{ideal}(\Gamma) \quad \text{and} \quad \text{cone}(\text{ideal}(\text{cone}(B))) = \text{cone}(B).$$

Since there are only a finite number of squarefree monomial ideals in S , it follows that there are a finite list of squarefree monomial ideals $B_1, \dots, B_s$ such that $\text{cone}(B_i)$ is full-dimensional and $\text{ideal}(\text{cone}(B_i)) = B_i$. We call these cones the **chambers** for S . By definition, the chambers are disjoint from one another.

Definition 3.2. To simplify notation, we enumerate the chambers as $\Gamma_1, \dots, \Gamma_s$ and write $B_1, \dots, B_s$ for the corresponding squarefree monomial ideal.

Remark 3.3. We will confirm below that each pair (S, B_i) determines a simplicial projective variety X_i . The rough statement is that X_i is the quotient $\text{Spec}(S) - V(B_i)/G$ (with the action of G induced by the grading group), but this is a “rough” statement because of various subtleties involving quotients of projective varieties. So there is a sense in which the multigraded ring S naturally corresponds to the set of toric varieties $X_1, \dots, X_r$.

This connection is even sharper from a homological perspective. The pair (S, B_i) determines a module category as follows: we write $\text{mod}(S)$ for the category of $\text{Cl}(S)$ -graded S -modules and we write $\text{mod}_{B_i}(S)$ for the subcategory modulo those modules that are annihilated by a power of B_i . To the pair (S, B_i) we can attach the quotient category $\text{mod}(S)/\text{mod}_{B_i}(S)$. This equals—essentially by definition—the category of sheaves on the quotient stack $[\text{Spec}(S) - V(B_i)/G]$ which we denote as $\mathcal{X}_i$. We will see later that $\mathcal{X}_i$ is closely related to the aforementioned toric variety X_i .

Corollary 3.4. *Each B_i is generated by products of r distinct variables.*

Proof. Let $x_{i_1} \cdots x_{i_p}$ be a minimal generator of B_i and assume $p > r$. We have that Γ_i is in the span of $\deg_{\mathbb{Q}}(x_{i_1}), \dots, \deg_{\mathbb{Q}}(x_{i_p})$. Since Γ_i is r -dimensional, after potentially reindexing, we can assume that $\deg_{\mathbb{Q}}(x_{i_1}), \dots, \deg_{\mathbb{Q}}(x_{i_r})$ are linearly independent. If Γ_i lies in the span of these, then $x_{i_1} \cdots x_{i_r}$ lies in B_i , contradicting the initial minimal generator assumption. But then some portion of Γ_i would correspond to an ideal containing $x_{i_1} \cdots x_{i_r}$, contradicting the fact that the chambers of squarefree monomial ideals must be disjoint. $\square$

Remark 3.5. By considering intersections of the chambers, we can endow $\text{Eff}(S)$ with a fan structure which is known as the secondary fan. However, for the applications to Cox categories and King’s Conjecture, we do not need the full structure of the secondary fan, we simply need to know the chambers. So we will omit a fuller discussion of the secondary fan.

Remark 3.6. Because each chamber is defined in terms of spanning sets of monomials, the hyperplanes used to define each chamber will have a special form. Namely, each such hyperplane will be spanned by a codimension 1 set of linearly independent vectors $\deg_{\mathbb{Q}}(x_i)$.

In particular, we can build a naive stratification of $\text{Eff}(X)$ as follows. Write r for the rank of $\text{Cl}(S) \otimes \mathbb{Q}$. For each linearly independent $(r - 1)$ -tuple of vectors $\deg_{\mathbb{Q}}(x_i)$, we get a hyperplane, and taking the union of all of these hyperplanes will slice $\text{Eff}(X)$ into a finite set of chambers, each of which belongs entirely to one of the chambers $\Gamma_1, \dots, \Gamma_s$. In particular, the chambers of the secondary fan are a coarsening of the chambers of this naive stratification.

We can describe the B_i in several equivalent ways. Recall that we have the map $\deg: \mathbb{Z}^n \rightarrow \text{Cl}(S)$; we write $\deg_{\mathbb{Q}}$ for the induced map to $\text{Cl}(S) \otimes \mathbb{Q}$.

Proposition 3.7. *Fix a chamber Γ_i for S and any ray $\rho \in \Gamma_i$. The following ideals are equal*

- (1) *The ideal B_i corresponding to Γ_i , i.e. the ideal B_{Γ_i} as in Definition 3.1.*
- (2) *The B_{red} radical of the ideal $\langle \mathbf{x}^{\mathbf{a}} \text{ such that } \deg_{\mathbb{Q}}(\mathbf{x}^{\mathbf{a}}) \in \Gamma_i \rangle$.*
- (3) *The ideal $\sqrt{B_{\rho}}$ defined as the radical of the ideal $\langle \mathbf{x}^{\mathbf{a}} \text{ such that } \deg_{\mathbb{Q}}(\mathbf{x}^{\mathbf{a}}) \in \rho \rangle$*

Proof. For (1) $\Rightarrow$ (3): Let $x_{i_1} \cdots x_{i_s}$ be a squarefree monomial which spans Γ_i . Since ρ lies in Γ_i , for some integers $a_1, \dots, a_s$ we have that $\deg_{\mathbb{Q}}(x_{i_1}^{a_1} \cdots x_{i_s}^{a_s}) \in \rho$. In particular, $x_{i_1} \cdots x_{i_s}$ belongs to $\sqrt{B_{\rho}}$.

Certainly the ideal from (3) belongs the ideal from (2).

Finally we have (2) $\Rightarrow$ (1): Let $x_{i_1}, \dots, x_{i_p}$ be the variables dividing $\mathbf{x}^{\mathbf{a}}$. Assume, for contradiction, that the corresponding degrees all line on some hyperplane H . Then H subdivides Γ_i into two pieces Γ_i^+ and Γ_i^- , corresponding to the two side of H . There must therefore be at least one monomial whose degree lies on either side of H , say $\deg_{\mathbb{Q}}(x_+)$ is on the Γ_i^+ -side and $\deg_{\mathbb{Q}}(x_-)$ is on the Γ_i^- -side. But then the ideals $B_{\Gamma_i^+}$ and Γ_i^- would be distinct, and so Γ_i would not be a valid chamber.

We conclude that $\deg_{\mathbb{Q}}(x_{i_1}), \dots, \deg_{\mathbb{Q}}(x_{i_p})$ span an r -dimensional cone. Since this span includes a point in Γ_i , by construction of the chambers, it must contain the entire chamber Γ_i . It follows that $x_{i_1} \cdots x_{i_p}$ lies in B_i . ♣♣♣ Daniel: [This is not well articulated yet, but it's definitely very doable.] $\square$

3.1. Chambers and toric varieties. From the irrelevant ideal B_i we can produce a fan Σ_i for a toric variety as follows. We let the rays of Σ_i be the set of rays ρ_j such that the corresponding principal ideal $\langle x_j \rangle$ does not contain B_i . And we let the cones of Σ_i be the cones spanned by the tuples $\rho_{j_1}, \dots, \rho_{j_p}$ such that B_i does not contain the monomial $x_{j_1} \cdots x_{j_p}$. The fans $\Sigma_1, \dots, \Sigma_s$ are all distinct because the B_i are all distinct.

For a point $\mathbf{a} = (a_1, \dots, a_n) \in \mathbb{Z}^n$ we define the polyhedron

$$P_{\mathbf{a}} := \cap_{i=1}^n \{m \in M \mid \alpha_i(m) \geq -a_i\}.$$

To a polyhedron $P_{\mathbf{a}} \subseteq M \otimes \mathbb{Q}$ there is a dual normal fan $\text{NormalFan}(P_{\mathbf{a}}) \subseteq N \otimes \mathbb{Q}$. We recall that $\text{NormalFan}(P_{\mathbf{a}}) \subseteq N \otimes \mathbb{Q}$ is defined as ...

The key fact relating these definitions is:

Proposition 3.8. *We have that $P_{\mathbf{a}}$ is nonempty if and only if $\deg_{\mathbb{Q}}(\mathbf{a})$ lies in the effective cone.*

Moreover, we have that $\deg_{\mathbb{Q}}(\mathbf{a}) \in \Gamma_i$ if and only if there is an open neighborhood $U_{\mathbf{a}}$ of $\mathbf{a}$ such that the normal fan of $P_{\mathbf{a}'}$ equals Σ_i for all $\mathbf{a}' \in U_{\mathbf{a}}$.

♣♣♣ Daniel: [To do: add references from CLS.]

Proposition 3.9. *As defined above we have the following: Σ_i is a simplicial fan such that $|\Sigma_i| = N \otimes \mathbb{Q}$. In particular, Σ_i is the fan associated to a simplicial, projective toric variety which we denote as X_i .*

Lemma 3.10. *The following are equivalent:*

- (1) *The Cox ring of X_i equals S ;*
- (2) *$\text{Cl}(X_i) = \text{Cl}(S)$;*
- (3) *The rays of Σ_i are precisely $\rho_1, \dots, \rho_n$;*
- (4) *B_i does not belong to any principal ideal $\langle x_j \rangle$.*

When these conditions are satisfied, the ample cone equals Γ_i . ♣♣♣ Daniel: [When these conditions are not satisfied, one needs to be a bit more careful in relating the ample cone with Γ_i .]

3.2. Irrelevant ideals for faces of secondary fan. ♣♣♣ Daniel: [Not sure what we want to put here.]

Lemma 3.11. *For a face of the secondary fan, the irrelevant ideal is the sum irrelevant ideals of the chambers with that face in its boundary.*